

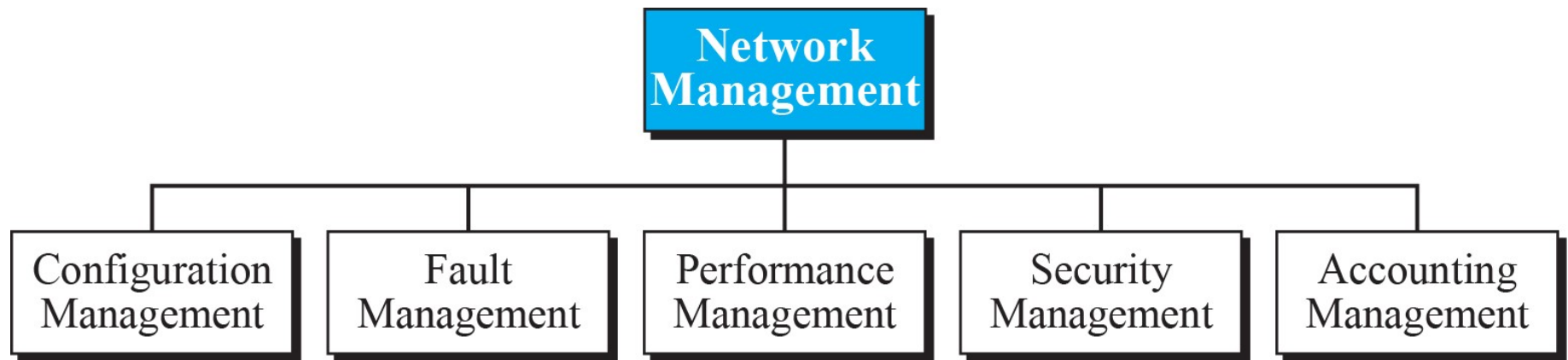
Simple Network Management Protocol (SNMP)



INTRODUCTION

- *We can define network management as monitoring, testing, configuring, and troubleshooting network components to meet a set of requirements defined by an organization.*
- *These include the smooth, efficient operation of the network that provides the predefined quality of service for users.*
- *To accomplish this task, a network management system uses hardware, software, and humans.*

Figure : *Areas of network management*



Configuration Management

- *A large network is usually made up of hundreds of entities that are physically or logically connected to each other. These entities have an initial configuration when the network is set up, but can change with time.*
- *Desktop computers may be replaced by others; application software may be updated to a newer version; and users may move from one group to another.*
- *The configuration management system must know, at any time, the status of each entity and its relation to other entities.*
- *Configuration management can be divided into two subsystems: reconfiguration and documentation.*

Fault Management

- *Complex networks today are made up of hundreds and sometimes thousands of components.*
- *Proper operation of the network depends on the proper operation of each component individually and in relation to each other.*
- *Fault management is the area of network management that handles this issue.*
- *An effective fault management system has two subsystems: reactive fault management and proactive fault management.*

Performance Management

- *Performance management, which is closely related to fault management, tries to monitor and control the network to ensure that it is running as efficiently as possible.*
- *Performance management tries to quantify performance using some measurable quantity, such as capacity, traffic, throughput, or response time. E.g. SNMP*

Security Management

- *Security management is responsible for controlling access to the network based on predefined policy.*
- *Security tools such as encryption and authentication. Encryption allows privacy for users; authentication forces the users to identify themselves.*

Accounting Management

- *Accounting management is the controlling of users' access to network resources through charges.*
- *Under accounting management, individual users, departments, divisions, or even projects are charged for the services they receive from the network.*
- *Charging does not necessarily mean cash transfer; it may mean debiting the departments or divisions for budgeting purposes.*

SNMP

- *Several network management standards have been devised during the last few decades. The most important one is Simple Network Management Protocol (SNMP), used by the Internet.*
- *SNMP is a framework for managing devices in an internet using the TCP/IP protocol suite.*

Managers and Agents

- *A management station, called a manager, is a host that runs the SNMP client program.*
- *A managed station, called an agent, is a router (or a host) that runs the SNMP server program. Management is achieved through simple interaction between a manager and an agent.*

Management Components

- *To do management tasks, SNMP uses two other protocols:*
- *Structure of Management Information (SMI) and Management Information Base (MIB).*
- *In other words, management on the Internet is done through the cooperation of three protocols: SNMP, SMI, and MIB.*

Management

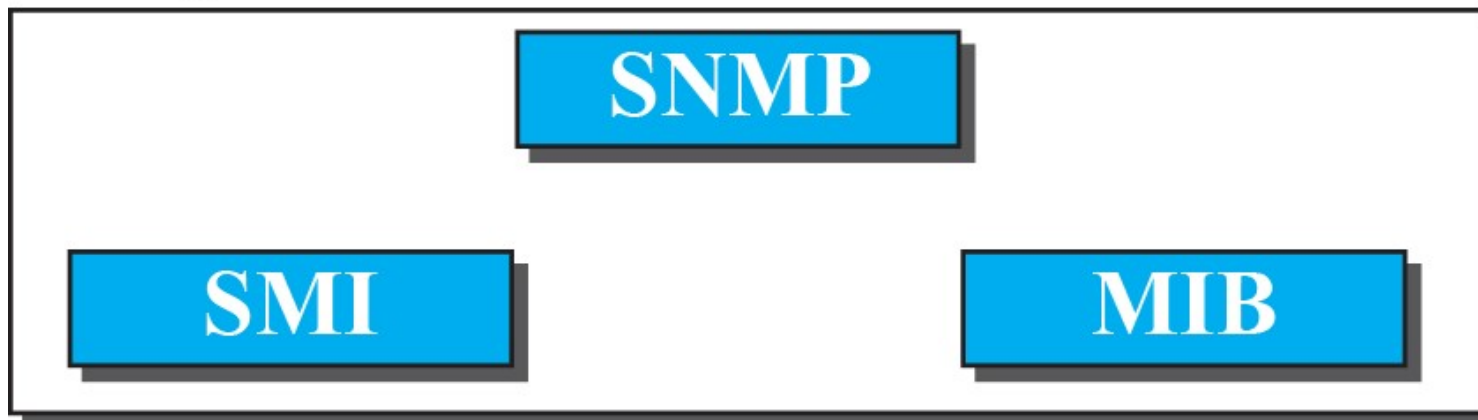
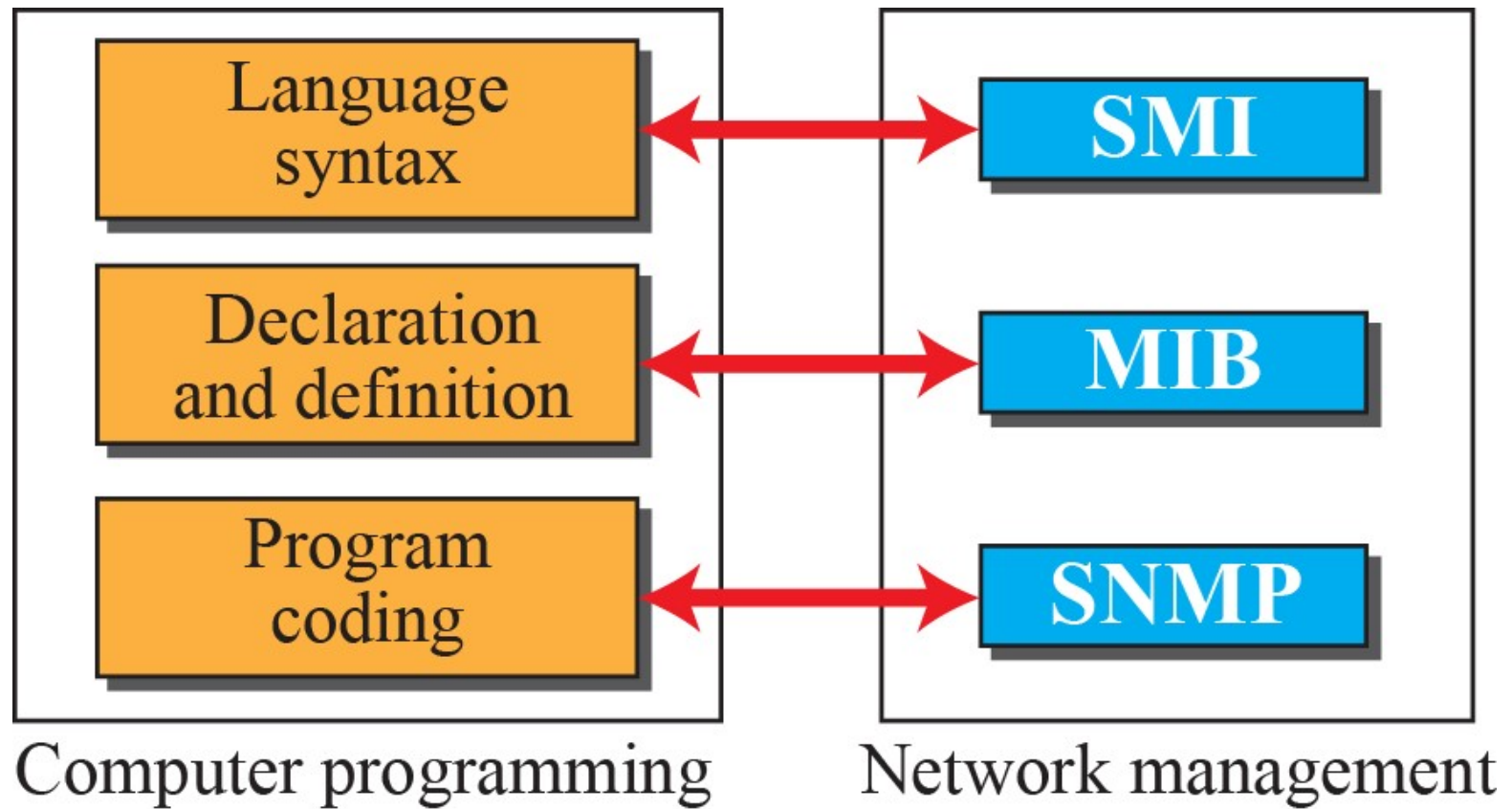


Figure : *Comparing computer programming and network management*

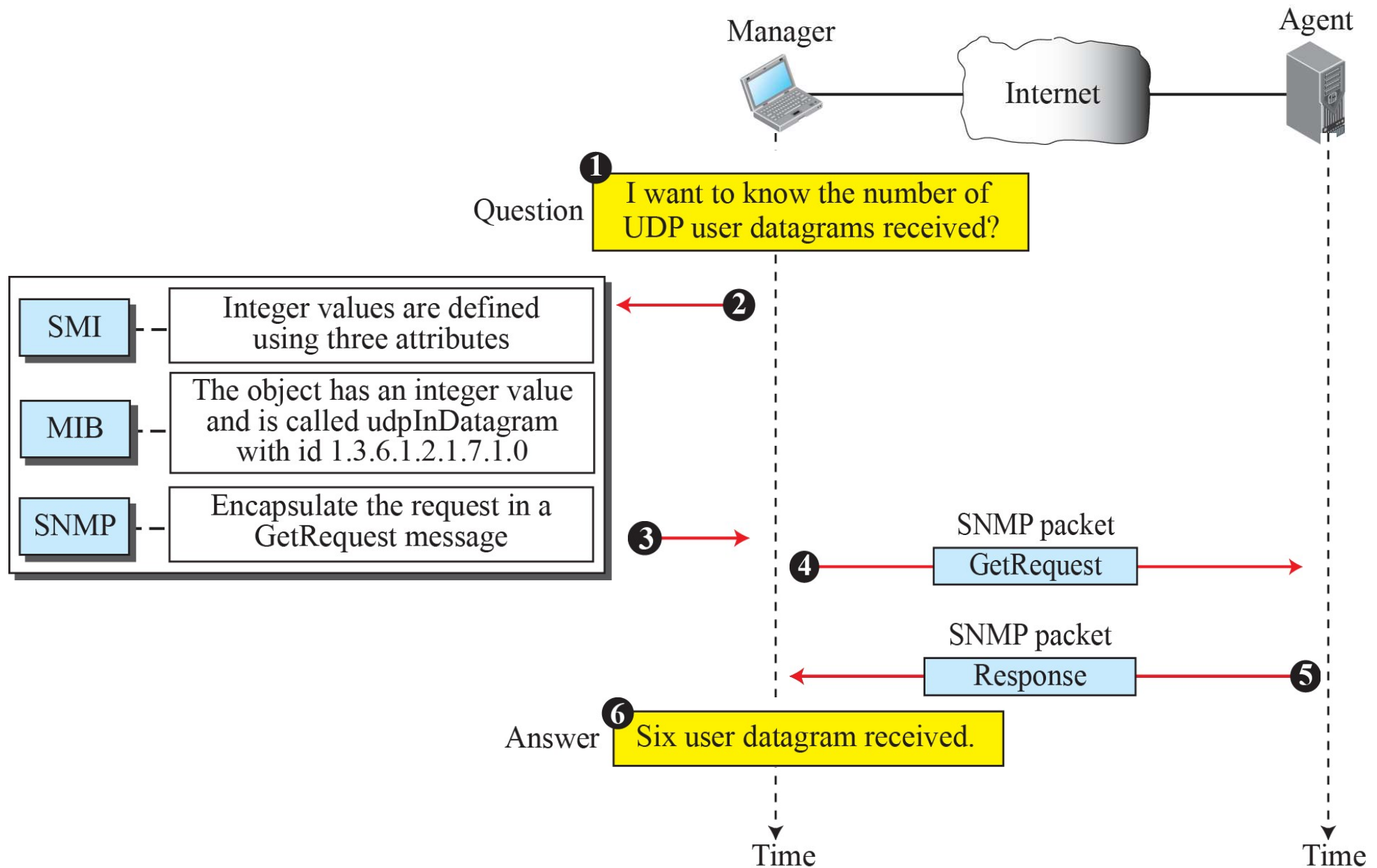


An Overview

Simple scenario:

A manager station (SNMP client) wants to send a message to an agent station (SNMP server) to find the number of UDP user datagrams received by the agent.

Figure : Management overview



SMI

The Structure of Management Information, version 2 (SMIv2) is a component for network management. SMI is a guideline for SNMP.

It emphasizes three attributes to handle an object: name, data type, and encoding method. Its functions are:

- *To name objects.*
- *To define the type of data.*
- *To show how to encode data.*

Figure : *Object identifier in SMI*

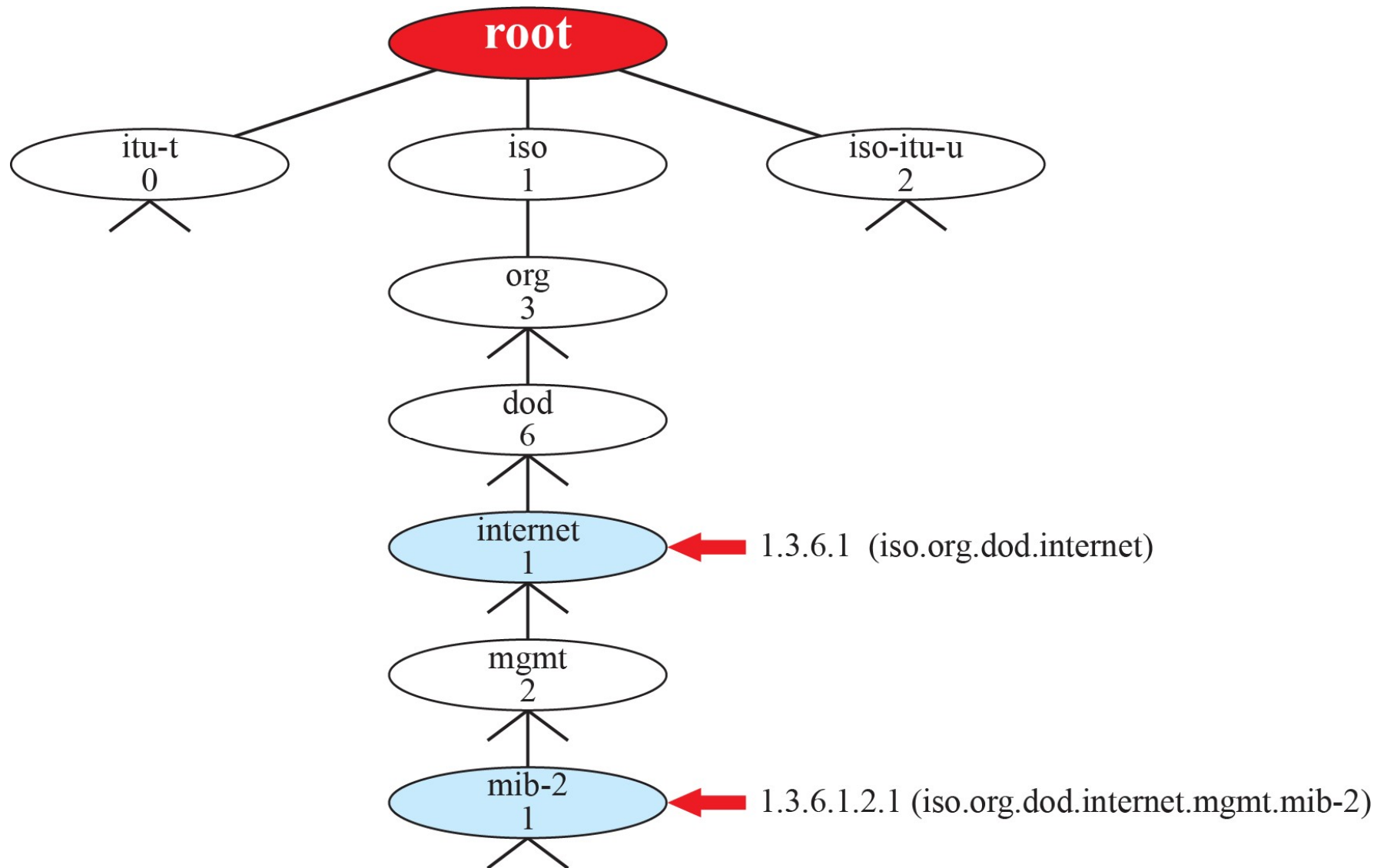


Table : Data types

<i>Type</i>	<i>Size</i>	<i>Description</i>
INTEGER	4 bytes	An integer with a value between -2^{31} and $2^{31}-1$
Integer32	4 bytes	Same as INTEGER
Unsigned32	4 bytes	Unsigned with a value between 0 and $2^{32}-1$
OCTET STRING	Variable	Byte-string up to 65,535 bytes long
OBJECT IDENTIFIER	Variable	An object identifier
IPAddress	4 bytes	An IP address made of four integers
Counter32	4 bytes	An integer whose value can be incremented from zero to 2^{32} ; when it reaches its maximum value it wraps back to zero
Counter64	8 bytes	64-bit counter
Gauge32	4 bytes	Same as Counter32, but when it reaches its maximum value, it does not wrap; it remains there until it is reset
TimeTicks	4 bytes	A counting value that records time in 1/100ths of a second
BITS		A string of bits
Opaque	Variable	Uninterpreted string

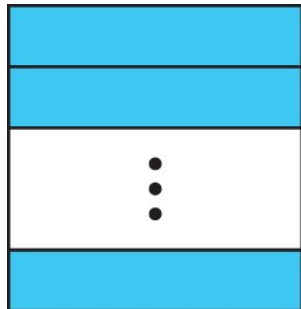
Figure : *Conceptual data types*



a. Simple variable



b. Sequence



c. Sequence of



d. Sequence of (sequences)

Figure : *Encoding format*

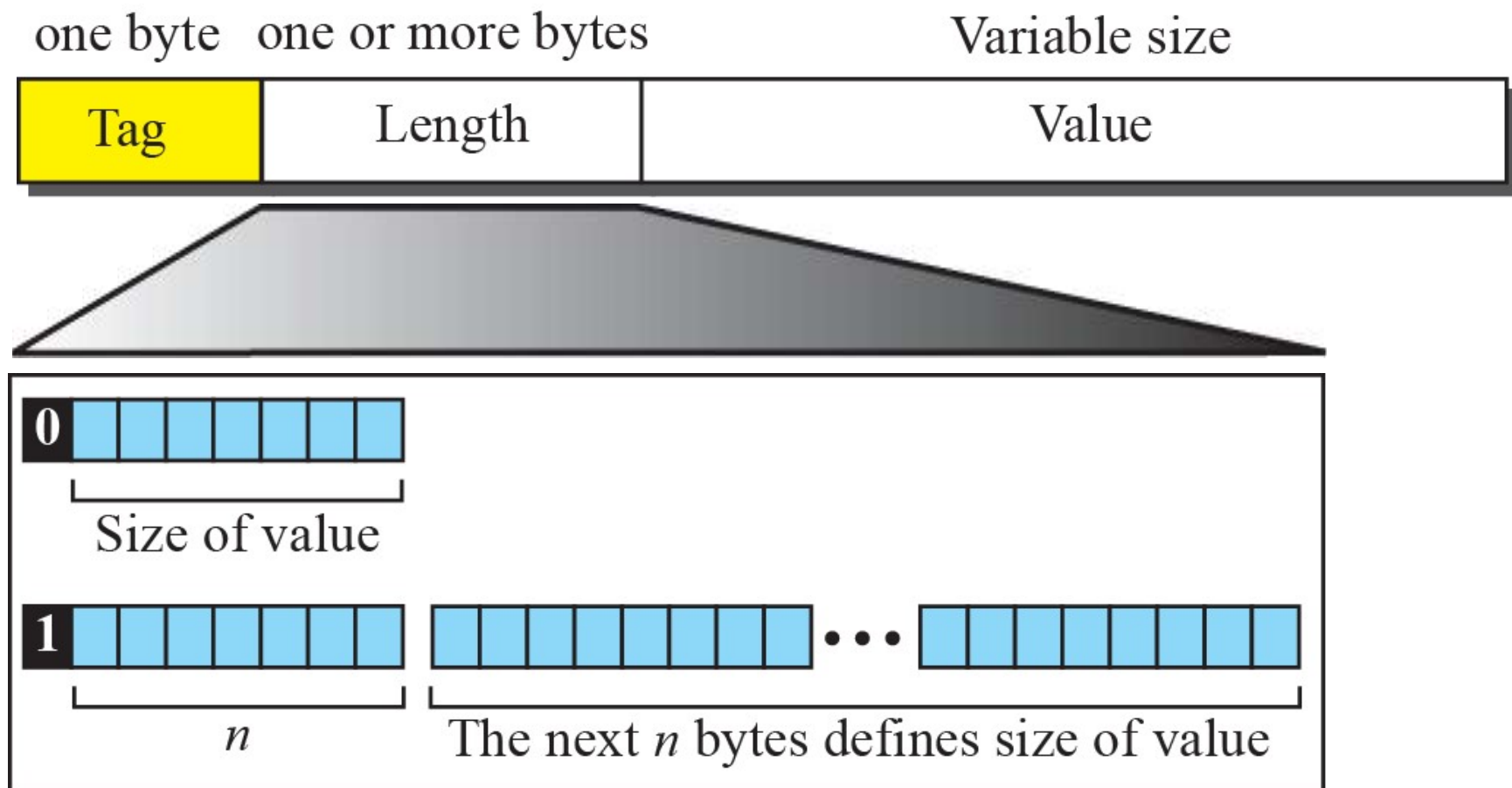


Table : Codes for data types

<i>Data Type</i>	<i>Tag (Hex)</i>	<i>Data Type</i>	<i>Tag (Hex)</i>
INTEGER	02	IPAddress	40
OCTET STRING	04	Counter	41
OBJECT IDENTIFIER	06	Gauge	42
NULL	05	TimeTicks	43
SEQUENCE, SEQUENCE OF	30	Opaque	44

Figure shows how to define INTEGER 14. The size of the length field is from Table 9.27.

Example : INTEGER 14

0x02	0x04	0x00	0x00	0x00	0x0E
Tag (integer)	Length (4 bytes)	Value (14)			

Figure shows how to define the OCTET STRING “HI.”

Example: OCTET STRING “HI”

0x04	0x02	0x48	0x49
Tag (String)	Length (2 bytes)	Value (H)	Value (I)

Figure shows how to define ObjectIdentifier 27.3.6.1 (iso.org.dod.internet).

Example : ObjectIdentifier 27.3.6.1

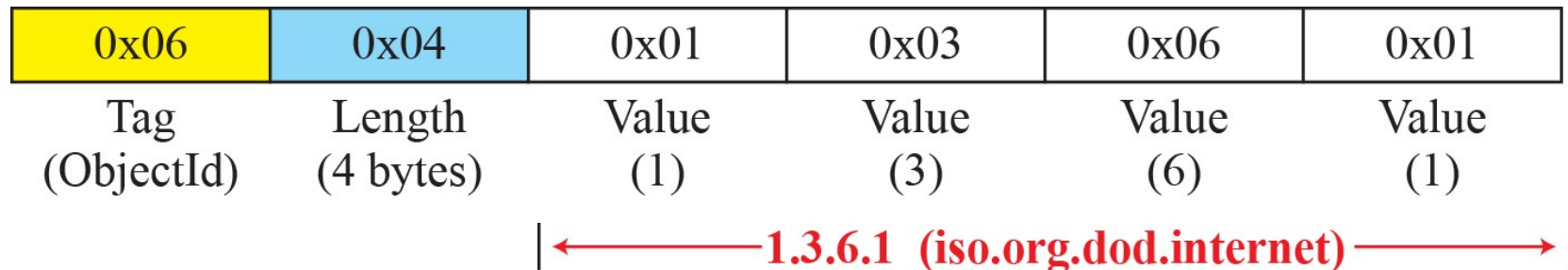


Figure shows how to define IPAddress 1327.227.14.8.

Example: IPAddress 1327.227.14.8

0x40	0x04	0x83	0x15	0x0E	0x08
Tag (IPAddress)	Length (4 bytes)	Value (131)	Value (21)	Value (14)	Value (8)

| ← 131.21.14.8 → |

The Management Information Base, version 2 (MIB2) is the second component used in network management. Each agent has its own MIB2, which is a collection of all the objects that the manager can manage. (See Figure.)

Figure: Some mib-2 groups

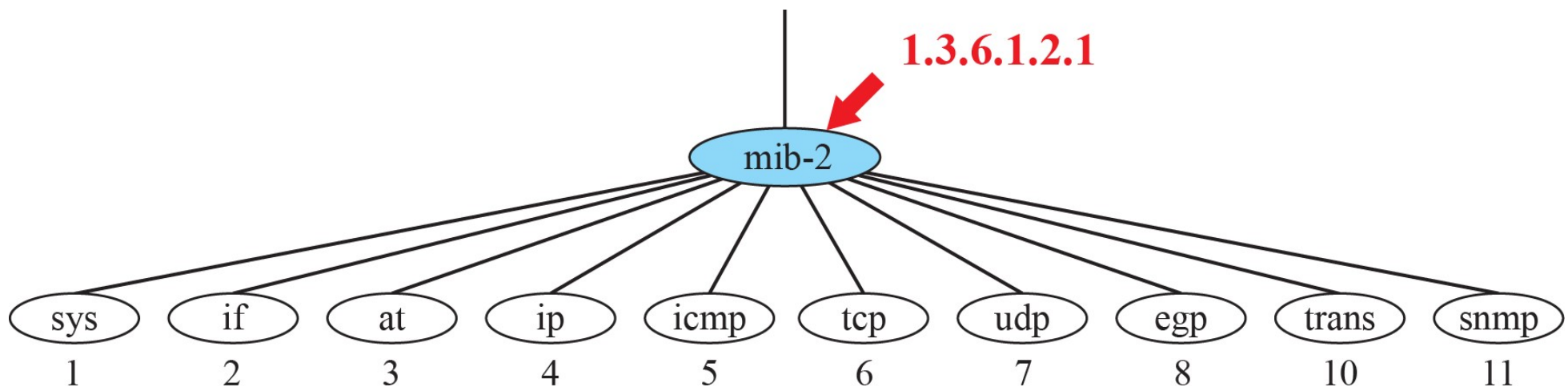


Figure : *udp group*

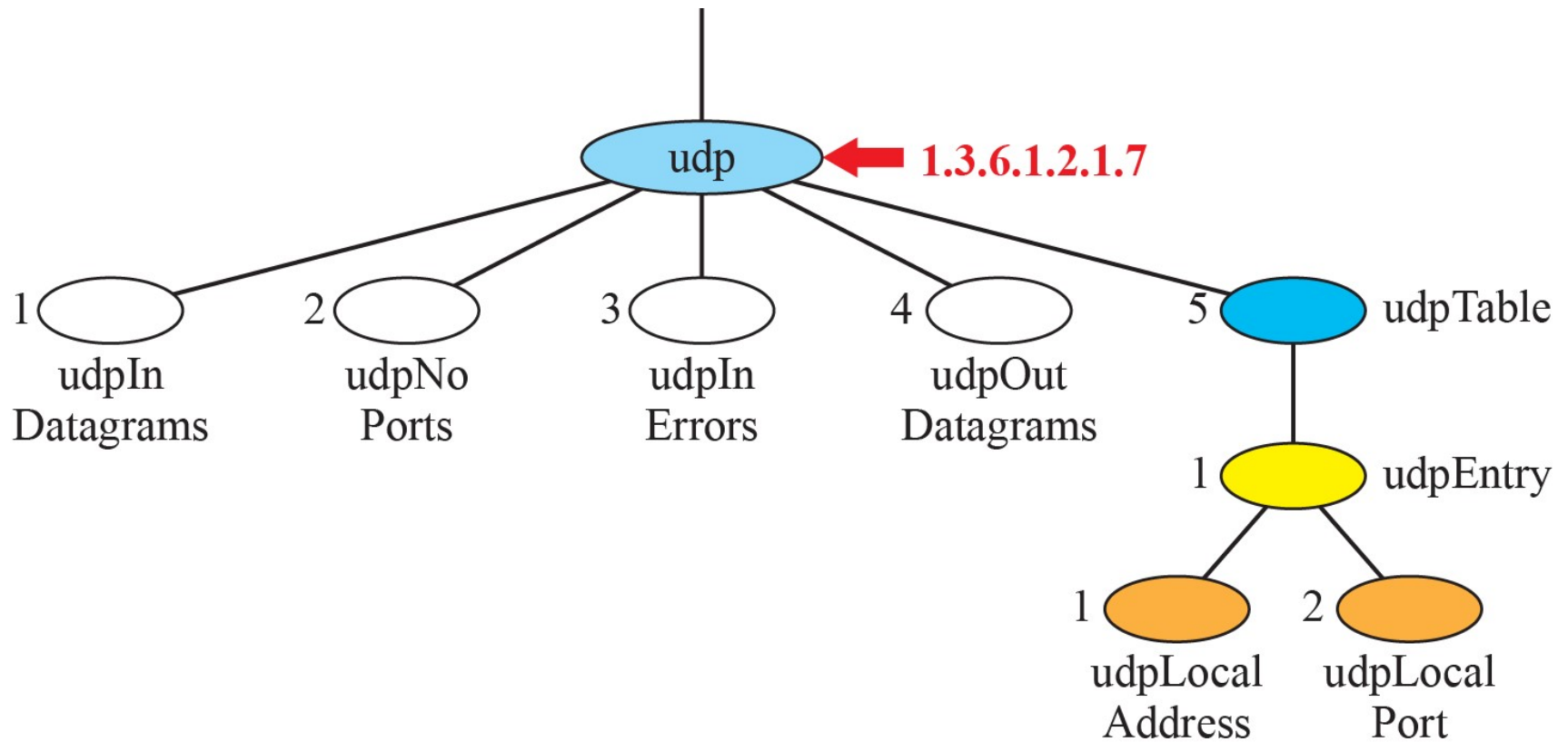


Figure : *udp variables and tables*


udpInDatagrams
(1.3.6.1.2.1.7.1)


udpNoPorts
(1.3.6.1.2.1.7.2)


udpInErrors
(1.3.6.1.2.1.7.3)


udpOutDatagrams
(1.3.6.1.2.1.7.4)

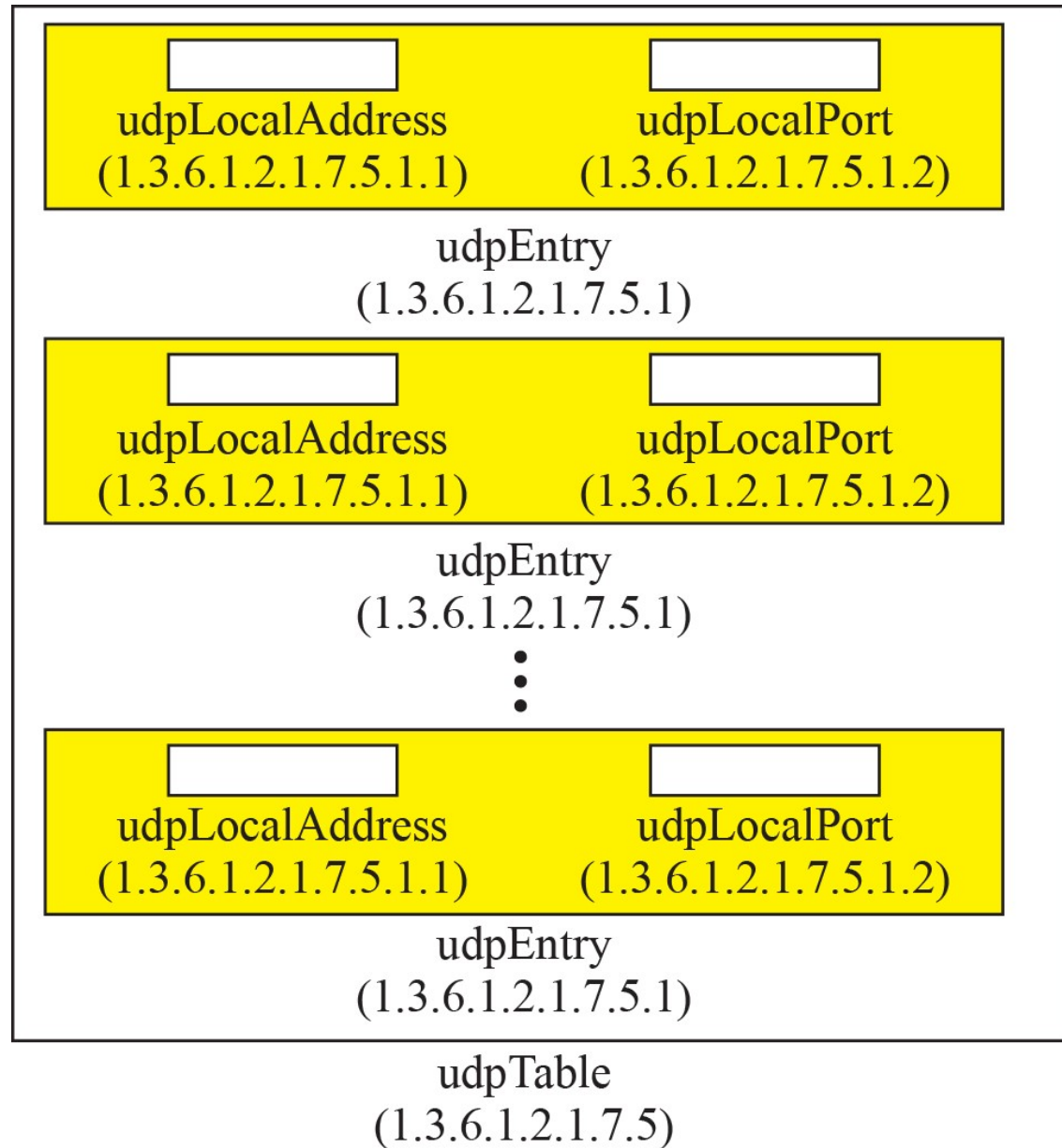


Figure : *Indexes for udpTable*

181.23.45.14 1.3.6.1.2.1.7.5.1.1.181.23.45.14.23	23 1.3.6.1.2.1.7.5.1.2.181.23.45.14.23
192.13.5.10 1.3.6.1.2.1.7.5.1.1.192.13.5.10.161	161 1.3.6.1.2.1.7.5.1.2.192.13.5.10.161
227.2.45.18 1.3.6.1.2.1.7.5.1.1.227.2.45.18.180	180 1.3.6.1.2.1.7.5.1.2.227.2.45.18.180
230.20.5.24 1.3.6.1.2.1.7.5.1.1.230.20.5.24.212	212 1.3.6.1.2.1.7.5.1.2.230.20.5.24.212

SNMP

SNMP uses both SMI and MIB in Internet network management. It is an application program that allows:

- A manager to retrieve the value of an object defined in an agent.*
- A manager to store a value in an object defined in an agent.*
- An agent to send an alarm message about an abnormal situation to the manager.*

Figure : SNMP PDUs

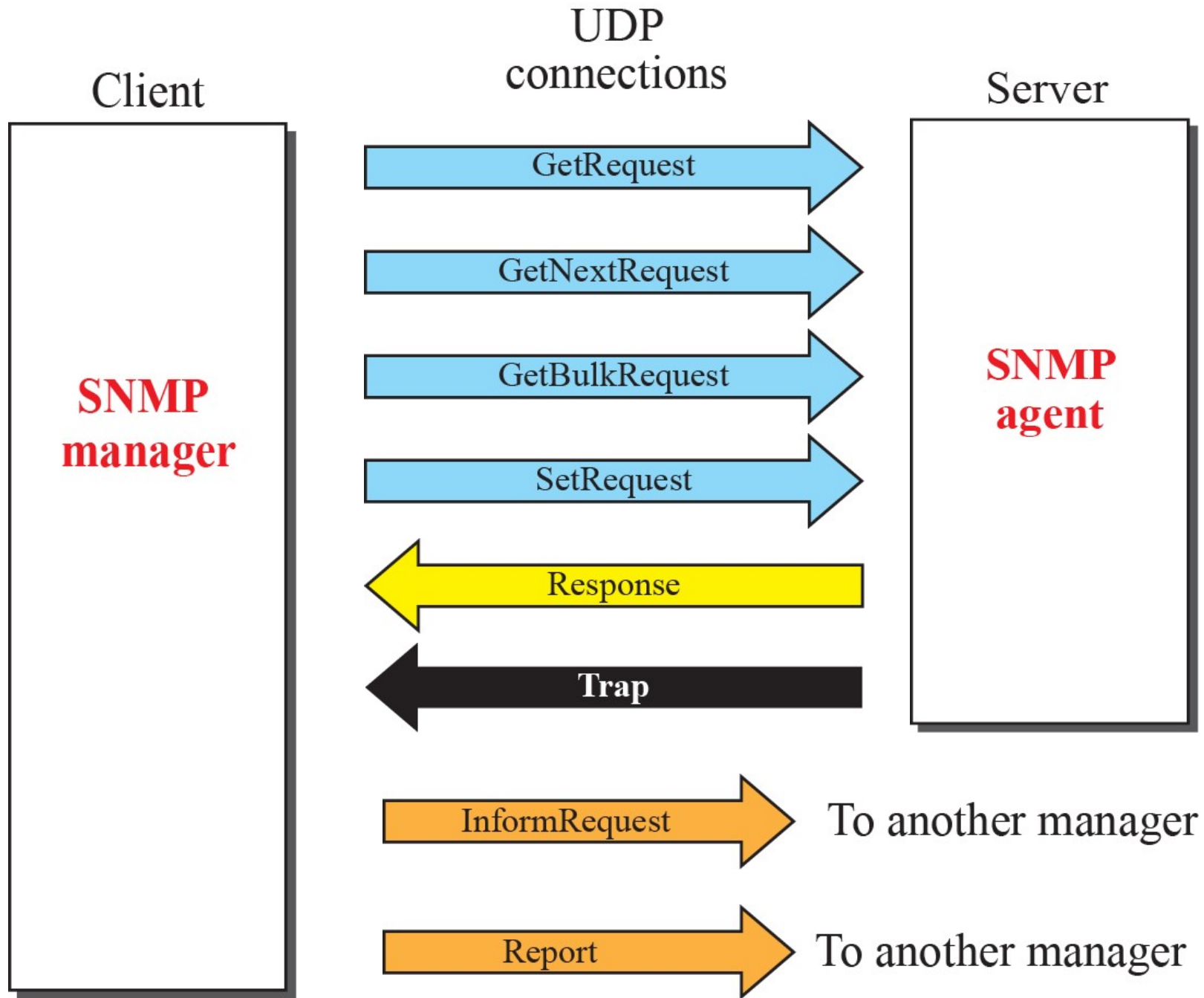
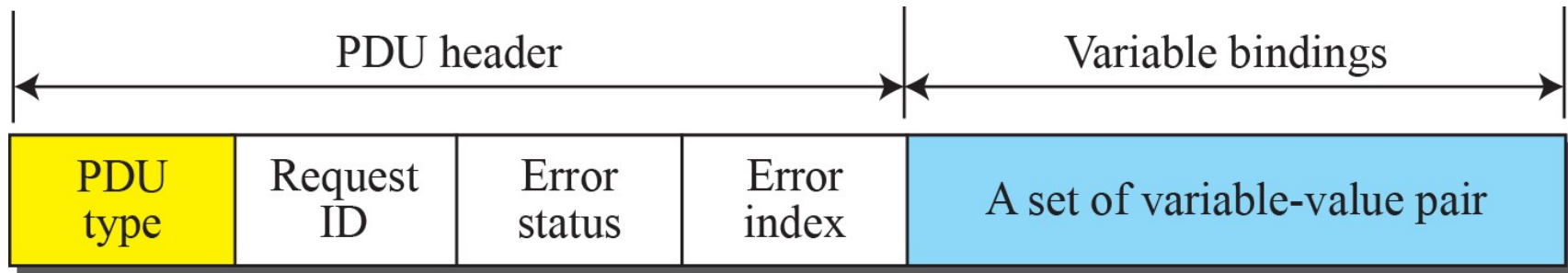


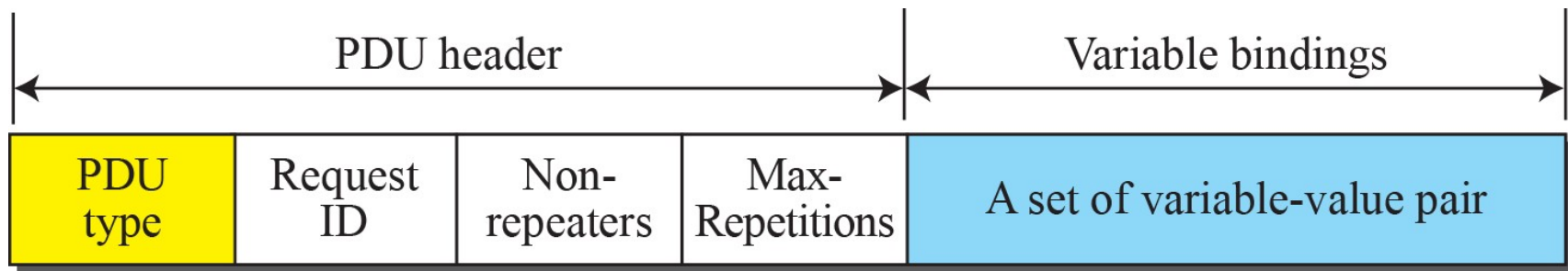
Figure : *SNMP PDU format*

Note:

The error status and error index values are set to 0 for all request messages.



a. All PDU types except GetBulkRequest



b. GetBulkRequest

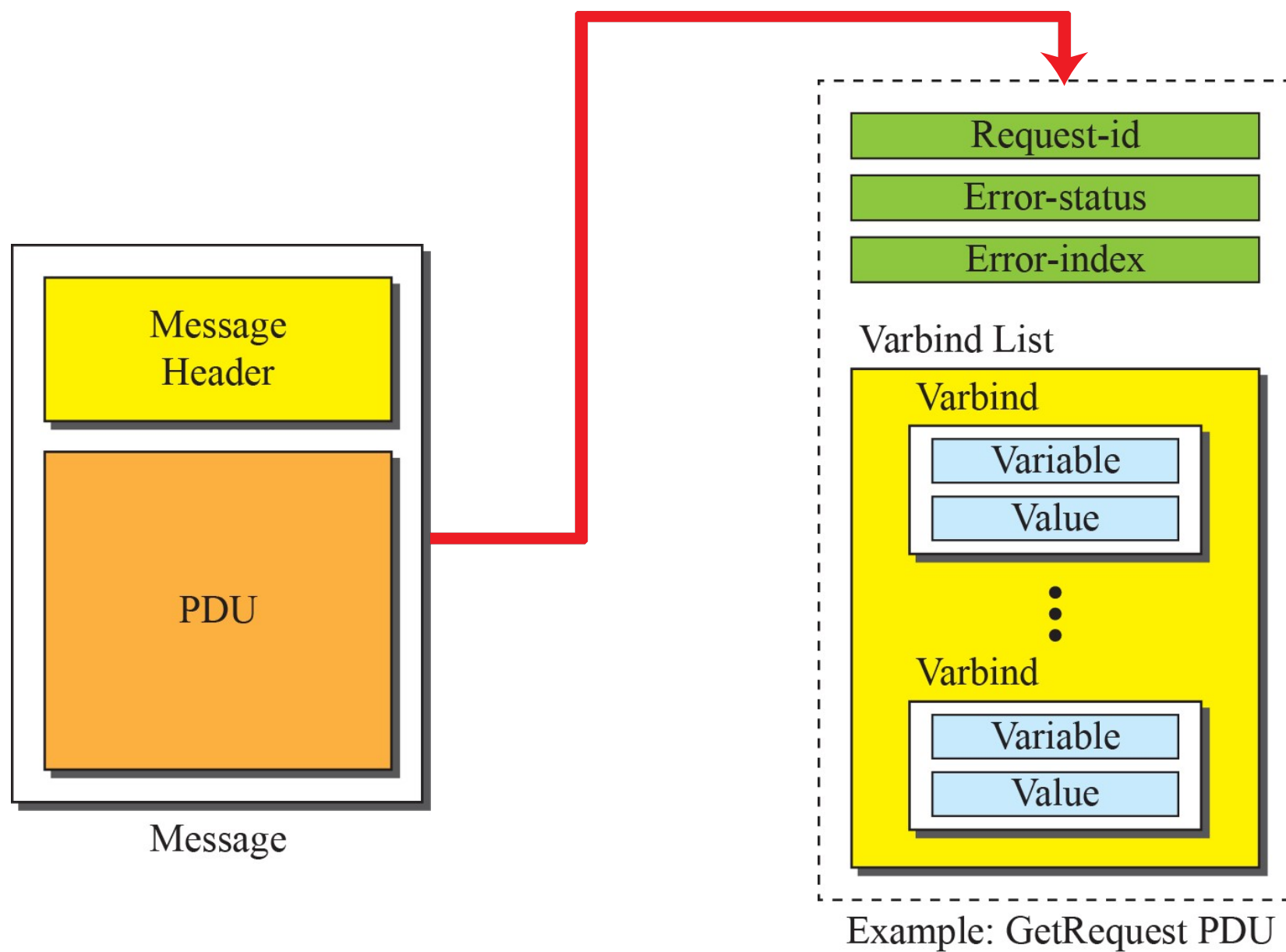
Table : PDU types

<i>Type</i>	<i>Tag (Hex)</i>	<i>Type</i>	<i>Tag (Hex)</i>
GetRequest	A0	GetBulkRequest	A5
GetNextRequest	A1	InformRequest	A6
Response	A2	Trap (SNMPv2)	A7
SetRequest	A3	Report	A8

Table : Types of errors

<i>Status</i>	<i>Name</i>	<i>Meaning</i>
0	noError	No error
1	tooBig	Response too big to fit in one message
2	noSuchName	Variable does not exist
3	badValue	The value to be stored is invalid
4	readOnly	The value cannot be modified
5	genErr	Other errors

Figure : *SNMP message*



Example

In this example, a manager station (SNMP client) uses a message with a GetRequest PDU to retrieve the number of UDP datagrams that a router has received (Figure).

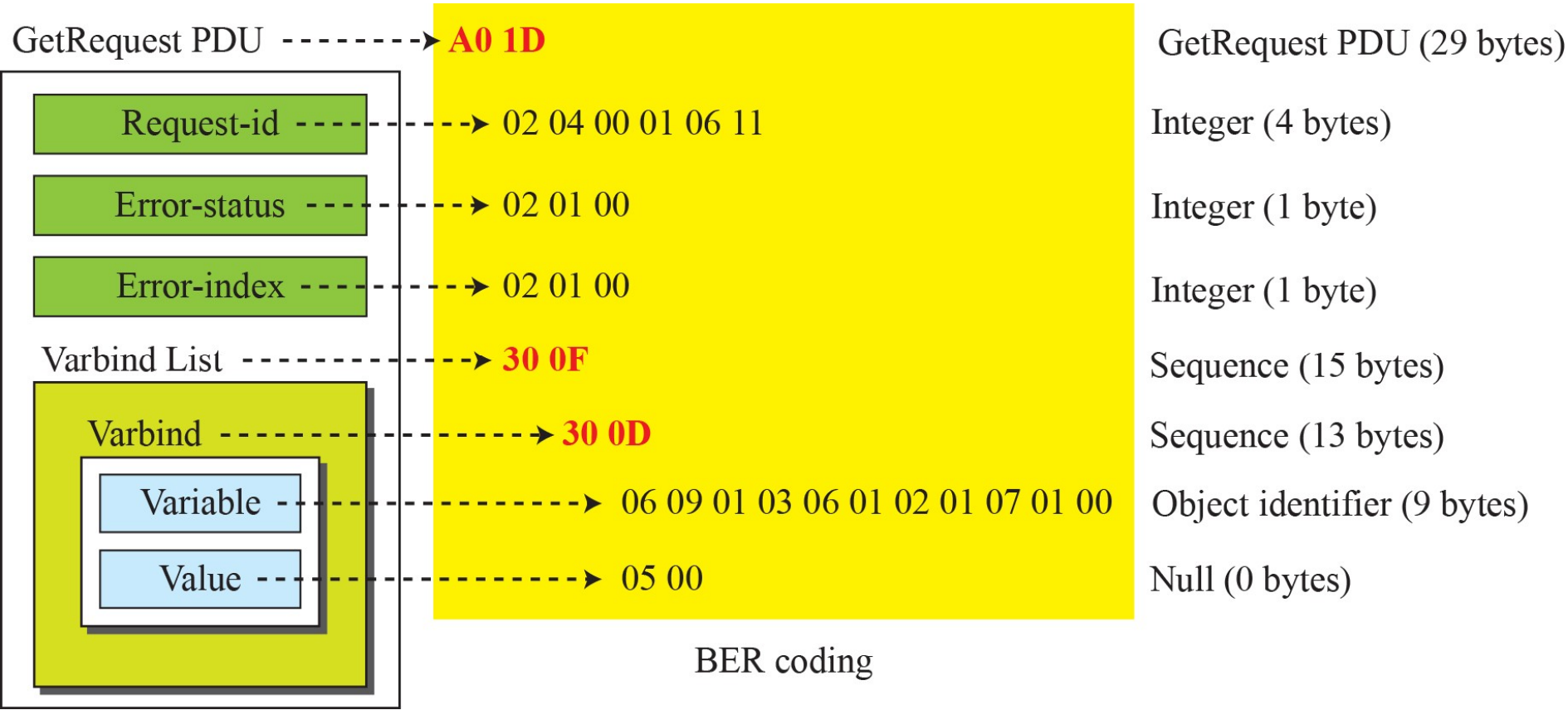


Figure : *Actual message sent for Example*

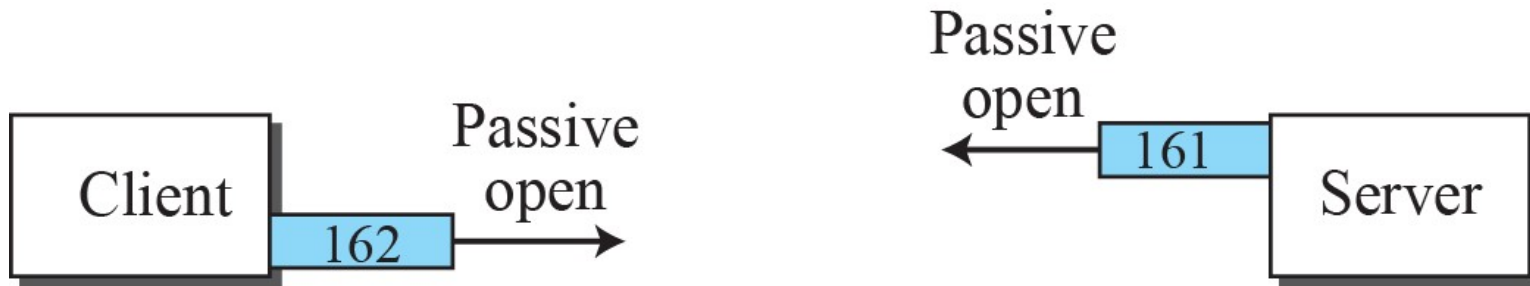
30	29	--	--
--	--	--	--
--	--	--	--
A0	1D	02	04
00	01	06	11
02	01	00	02
01	00	30	0F
30	0D	06	09
01	03	06	01
02	01	07	01
00	05	00	

Message

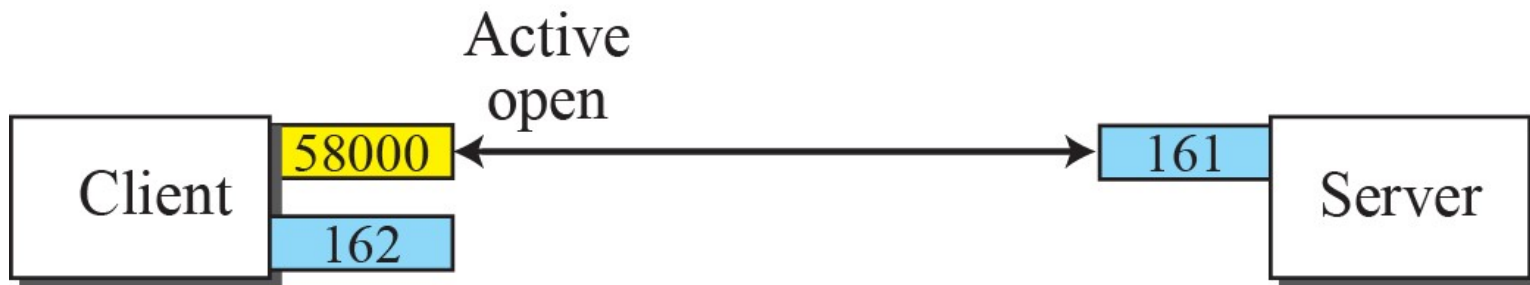
Note:

The byte values
are in hexadecimal

Figure : *Port numbers for SNMP*



a. Passive open by both client and server



b. Exchange of request and response messages



c. Server sends Trap message

Thank You...